POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Chemistry

Semester: Fall

Year : 2017
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Derive Henderson-Hassel Balch equation for Acidic buffer. What is the pH of 0.50M aqueous NaCN? Pkb for Cynide ion is 4.70. 8
   b) Define electrochemical series with applications. Also explain rusting of iron with mechanism. 7
2. a) Define ionization energy. Why second ionization energy is always greater than first ionisation energy? Explain with suitable reason and examples. 7
   b) Give reasons 8
      i. Transition elements show paramagnetism
      ii. $[Mn(OH)_6]^{2+}$ is pale pink, MnO_2 is black and MNO_4^- is intensely purple colour
3. a) Define free radicals with formation, structure and stability. 7
   b) Differentiate SN1 reaction and SN2 reaction in terms of their mechanism. 8
4. a) Write notes on vulcanization, Nylon and Teflon. 7
   b) Describe the mechanism of condensation and addition polymerization reaction. 8
5. a) Describe the principle of Nuclear Magnetic spectroscopy. 5
   b) Draw a neat and labled diagram of Mass Spectrometer. 5
   c) Describe the chemistry of cement. 5
6. a) Define Ozone layer with its importance. 5
   b) Show your aquitance with global warming and green house effect. 5
   c) Write various causes of water pollution. How it can be control. 5
7. Write short notes on: **(Any two)** 2×5
   a) Chromatography
   b) Chemical oxygen demand
   c) Optical activity

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